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Effects of personalization and social role in voice shopping: An experimental study on product recommendation by a conversational voice agent

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ABSTRACT

This study examines the persuasion mechanism in product recommendations made by a voice-based conversational agent and explores whether the personalized content reflecting the customer's preferences and the agent's social role of a friend, rather than a secretary, generate a more positive attitude toward the product in the context of voice shopping. With the framework of dual modes of information-processing models, we hypothesized that the personalization of messages would be a central route with a greater impact on attitude for products with high involvement. By contrast, the social role of the conversational agent was expected to represent a peripheral route with a greater impact on products with low involvement. An experimental study was designed to test the effects of personalized content that reflected individual preferences for product attributes and a friend role of a voice agent with high and low product involvement. The results showed main effects of both personalization and the social role on building attitudes toward the product. Although no interaction effect for personalization and involvement was found, there was a significant interaction effect for the social role and involvement. This study contributes to persuasion theory by extending it to the interaction with a conversational agent. For practitioners, the study provides insights into the importance of the personalized content of recommendations and the need for consideration of an alternative social role in the design of voice shopping through a conversational agent.

1. Introduction

Alexa, OK Google, Siri, Cortana—these names are becoming increasingly familiar. People rely on these conversational voice agents to obtain information such as today's weather and news, or ask them to do things on their behalf, such as play music, turn on the lights, or send text messages. As technology evolves, these voice agents powered by artificial intelligence (AI) are being integrated into diverse devices, from computers and smartphones to Bluetooth speakers and connected home appliances such as refrigerators or televisions.

Often, these conversational agents are positioned and promoted as personal assistants, who give advice based on the information to which agents have access. One type of advice that conversational agents provide is a product recommendation. As product recommendations by agents are widely implemented in various devices, market research has estimated that the voice-shopping market will grow to USD 40 billion in 2022 (Perez, 2018). Industry and marketers are starting to prepare for the emerging voice shopping by paying more attention to voice search engines and how to optimize content and products for them (Bentahar, 2018).

E-commerce has extended usability engineering to integrated designs for persuasion (Nah & Davis, 2002). Research on product recommendation has also shifted from a technological focus on algorithms (Bonhard & Sasse, 2006; Sinha & Swearingen, 2001) to social and psychological aspects (Chen & Lee, 2008; Sher & Lee, 2009). Until now, most studies on persuasive agents or recommendation agents have focused on the design of websites or text-based chatbots (Araujo, 2018; Hill, Ford, & Farreras, 2015; Huang & Zhou, 2018), but recently, voice interaction with a conversational agent, which usually has no visual display, has emerged as a new challenge for human–computer interaction (HCI) research and e-commerce.

Although textual conversations with a chatbot or artificial agent have been a popular subject in HCI research (Hill, Ford, & Farreras, 2015; Von der Pütten, Krämer, Gratch, & Kang, 2010), studies on voice interaction with an AI-powered conversational agent have been rare. Product recommendation by a conversational agent via a smart speaker

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or other devices can be considered a modified version of advertisement because they share the same goal: promote and sell a product or service. Easy, convenient access via voice interaction in a customer's home is one of the most attractive features of the conversational agent for shopping.

However, voice interaction with an agent for product recommendation has considerable limitations in terms of quantity, structure, and processing time of information (Carmel, 2019; Mari, 2019). In contrast to websites or mobile applications, conversational voice agents can deliver less information, with no visual or textual content. Moreover, in a turn-taking conversation, it is difficult to allow more than two or three choices for product comparison. For these reasons, it is challenging to fully promote a product and build positive attitudes toward the recommended product. Thus, to facilitate better user experience and create designs that are more persuasive for voice shopping, we should investigate the design strategies of the conversational agent and the message.

A personalized message for recommendations has been widely used as a promotional strategy in marketing, and research results have shown that personalization of advertising messages on the web is a strong determinant of building positive attitudes toward the product and brand (Ho & Bodoff, 2014; Kang, Shin, & Kong, 2016; Tam & Ho, 2005). Empowered by technological advances in big data analysis, machine learning, and natural language processing, online recommendation agents will evolve to better support each consumer's shopping decisions by eliciting individual preferences for product attributes (Ghasemaghaei, Hassanein, & Benbasat, 2019). Although the rise of voice-based recommendation agents seems to provide new perspectives on e-commerce, research has not addressed the question of how and to what extent text-based personalization in web and mobile applications can be employed to improve the voice-shopping experience. Given the limitations of voice-only interaction with the conversational agent, the effect of personalization in voice shopping must be re-examined and validated to determine whether it is still a good strategy under the condition of limited information for a product recommendation.

Additionally, it is critical to explore a new persuasive strategy for the conversational agent. For interactive marketing campaigns on social media, most brands now adopt a more informal, personal, and closer tone of voice, unlike the formal corporate voice (Barcelos, Dantas, & Sénécal, 2018). As companies are paying more attention to building a close relationship and engagement with customers on interactive marketing platforms, the impact of an alternative social role, such as a friend, should be examined as a key characteristic of the recommendation agent. Although most current conversational agents assume a social role of a personal assistant or a secretary, voice shopping usually takes place at home for buying products of daily use. As people often rely on their friends' recommendations for product purchases when information is limited, the question as to what would occur if the agent has a different social role of a friend instead of a secretary appears relevant.

Another important research question relates to the mediating effect of product type. Given the expansion of shopping experiences with the agent to diverse product categories, whether the product type or the level of involvement with the product mediates the routes of a persuasive strategy for voice shopping should also be investigated. In addition, to deepen the understanding of the persuasion process, an examination of the possible interaction effects among personalization, social role, and product involvement is necessary. In many cases, the type or value of a product that a shopper wants to purchase determines the level of involvement and the cognitive strategy in attitude formation (Petty, Cacioppo, & Schumann, 1983; Zaichkowsky, 1985). For example, expensive products or search goods such as a car or a smartphone require a high level of involvement and use the central route of the persuasion process. By contrast, low-cost products or experiential goods such as wine or a movie generate a low level of involvement, and a peripheral route of the persuasion process is used.

We conducted an experimental study with a set of research hypotheses on the personalization of a message, the social role of an agent, and the level of involvement with a product. The main question that this study attempted to answer relates to how a personalized message and an agent's social role affect attitudes toward the product, according to the level of product involvement in the context of voice shopping.

2. Literature review

2.1. Dual-process models of persuasion and involvement with the product

How does the conversational voice agent affect the persuasive process to build a positive attitude toward the recommended product? The process of attitude change can be explained by dual-process models of persuasion. The two most prominent, traditional theories are the elaboration likelihood model (ELM) and the heuristic-systematic model (HSM): both posit that attitude change operates through two modes of persuasive message processing.

The ELM explains how attitude formation occurs when a person is exposed to persuasive messages and suggests that the persuasion process occurs differently based on the person's likelihood to elaborate on messages (Petty & Cacioppo, 1986, 2012). The extent to which the person will devote cognitive effort is referred to as the elaboration likelihood; elaboration involves careful processing and thorough evaluation of the argument, which requires cognitive effort. The central route is characterized by the amount of cognitive effort used in processing information. This route requires a person to think critically about the arguments presented in the message and evaluate and scrutinize the merits and details of the related information. As the person performs a thorough review of the arguments, the quality of the arguments has a critical impact on any change in their attitude (Bhattacherjee & Sanford, 2006; Sussman & Siegal, 2003). The argument quality refers to content directly related to the issue or the product (Areni & Lutz, 1988). By contrast, peripheral routes require less cognitive effort (Petty & Cacioppo, 1986) and involve the processing of information without critical evaluation of the message. Attitude formation occurs based on relatively simple heuristic cues and inferences such as characteristics of a recommender (Li, 2013; Yang, Hung, Sung, & Farn, 2006).

The HSM (Chaiken & Ledgerwood, 2011; Eagly & Chaiken, 1993; Todorov, Chaiken, & Henderson, 2002) shares the dual modes of persuasive message processing with the ELM. The level of cognitive ability and capacity is a critical trigger of different modes of information processing and attitude change. In systematic processing, people tend to scrutinize the message and form their attitude based on the quality of recommendation content. In heuristic processing, people are likely to exert less cognitive effort to evaluate the quality of content and form their attitude based on other communication cues (Chen & Chaiken, 1999).

In both models, the level of involvement with the product is a key factor that determines the processing mode of central/systematic or peripheral/heuristic route. Zaichkowsky (1985) defined product involvement as "a person's perceived relevance of the object based on inherent needs, values, and interests." Personal relevance of the persuasive message or motivation is a critical trigger of involvement (Chaiken & Ledgerwood, 2011; Cho, 1999; Eagly & Chaiken, 1993; Petty & Cacioppo, 1986). If a customer feels a strong motivation for a product, the message recipient will be more involved with the recommended product and more likely to invest cognitive effort to evaluate the message. By contrast, if motivation is low, the person will tend to have a low-involvement level. In a low-involvement state, people rely more on simple peripheral cues, such as the attractiveness of the endorser.

Petty et al. (1983) investigated how attitude formation occurs when a person is exposed to a personally relevant advertisement. They manipulated the level of motivation for the product by telling one group that the product would soon be available in their region, and by telling another group that the product did not have a release plan in their region. The results showed that in the high-involvement group, the quality

of the argument had a greater influence on attitude change, but in the low-involvement group, the attractiveness of the model in the advertisement had a greater impact. Thus, motivation determines the level of involvement, which moderates the effects of argument quality and heuristic cues on attitude change.

2.2. Computers as social actors and social role theory

How does a user perceive the virtual agent as a social actor? Reeves and Nass (1996) investigated how human–computer interaction works and found evidence of social responses when users interact with computers. Users tended to be polite to the computer, implied gender stereotypes, and reacted to the same voice from different machines as a single social presence. The researchers concluded that users recognized the agents in the computer they interacted with as "social actors" and applied social norms or social categories, such as gender or social role, as though they were interacting with human beings. Based on the CASA paradigm, the use of minimal social cues in HCI is expected to engender similar social behaviors and interactions to those that are observed in real life (Nass, Moon, & Green, 1997; Reeves & Nass, 1996). Many studies have been conducted on the CASA paradigm, and its applications have been expanded from computers to other technological artifacts such as chatbots, pedagogical agents, and conversation agents as technology has evolved (Baylor & Kim, 2004; Kim & Wei, 2011).

When investigating the social traits of the conversational agent, one of the key cues that trigger a social response is the social role imputed to the agent. According to social role theory, a role refers to the set of shared, normative expectations that can explain the behaviors of a person with a social position within a society (Biddle, 1986). A social role includes rights, duties, norms, expectations, and behaviors of an individual in a given social position. When a person is asked a question, the tone or content of their reply may vary depending on who else is participating in the conversation (Prendinger & Ishizuka, 2001).

When interacting with a conversational agent, users use communication cues related to social roles to define the agent and the social context of the relationship between the user and agent. Thus, what would be an effective social role for an AI-assisted conversational voice agent recommending a product to a user at home? As there are expectations of behaviors on the basis of the role an individual plays in a social structure, the tasks and social roles that will be performed must be considered when designing a conversational agent. If an agent has a specific social role assigned, the user will be able to set the appropriate expectations and behave accordingly, which enhances the effectiveness of the interaction and the overall user satisfaction with the agent (Isbister & Hayes-Roth, 1998). Currently, most conversational agents have the social role of a secretary or a personal assistant. As the secretary persona has a formal speech style and social norms, a friend persona could be an alternative social role, with an informal speech style, less psychological distance, and more similarity to the user.

3. Research model and hypotheses

Based on the review of dual processing models of persuasion and social role theory, this study developed a set of hypotheses to identify the effects of personalization, social role, and product involvement on the building of a positive attitude when a conversational voice agent recommends a product. Personalized recommendation of the message and the social role of the agent were adopted as central and peripheral routes, and involvement with the product was set as a mediating condition.

3.1. Personalization

The fundamental idea of personalization is to treat each customer as a unique entity and design the customized recommendation message based on his or her preferences (Kalyanaraman & Sundar, 2006). Thus, the design of a personalized message is to create a match between the customer's preferences and the content of the recommendation message. A personalized message that matches the customer's preferences is expected to be more persuasive than a standardized message that mismatches.

In e-commerce research, the dual processing of information models has been used to comprehend the manner in which the attitude toward the product is formed, and whether personalization is an efficacious strategy in generating a more positive attitude and purchase intention. In ELM, the central route involves the quality of the argument. The persuasiveness of a message is determined by how well it elucidates the merits of the argument in a detailed manner. In a product recommendation context, if the recommendation matches the user's tastes and preferences, the user is more likely to assume that the argument is strong, and develop a favorable attitude (Ho & Bodoff, 2014; Tam & Ho, 2005). Sher and Lee (2009) used the ELM to explore the effects of online reviews on purchase intention. The quality of the arguments in product reviews and the number of reviews influenced purchase intention through central and peripheral routes. To investigate the effects of web-based personalization strategies, Tam and Ho (2005) conducted an experiment using a website for ringtone downloads. Along with a recommendation set size and customized sorting order, the level of preference matching was found to be a key variable.

However, other studies have not demonstrated a significant difference between personalized and non-personalized messages in terms of the persuasion effect (Li, 2016; Noar, Benac, & Harris, 2007). Li and Liu (2017) argued that the effect of personalization was inconsistent and confusing in the literature, and this confusion might be generated because there were multiple approaches to matching messages with a customer's characteristics and preferences. For example, incorporating a customer's name in the message was a widely adopted approach to personalization. However, this approach to a personalized greeting may not be a stronger strategy than creating customized content for the message by matching a customer's preferences to specific attributes of a product, or a customer may perceive the personalized greeting as a heuristic cue when it is not directly relevant to the content of the recommendation message. They suggested that a more sophisticated personalization approach is required.

With recent technological advances in the utilization of usage history of a customer and collaborative filtering, personalized product offerings and tailored recommendation messages have been widely implemented on e-commerce sites (Zanker, Ricci, Jannach, & Terveen, 2010). We hypothesize that the personalized content of product recommendation messages is stronger than product recommendations made without personalization.

H1. Recommendations with personalized content in voice shopping will lead to a more positive attitude toward the product than recommendations with non-personalized content.

3.2. Social role of conversational agent and informal language style

There are many social cues such as gender or personality traits that a user may notice while interacting with a conversational agent. In the field of HCI, the design of conversational agent personas has mainly focused on gender as a social attribute associated with user motivation and interaction style (Benbasat, 2010; Brahnam & De Angeli, 2012). However, the social role of the agent has been rarely studied as an attribute assigned to the agent. Social role is a critical factor because it sets expectations regarding how the agent will behave according to the social norm of the role, and it constrains users' behavior based on the social status and structure of the relationship (Biddle, 1986).

Most conversational agents implicitly have the social role of secretary, for example, commonly used voice agents such as Siri and Alexa are promoted as personal assistants. Sinha and Swearingen (2001) compared preferences for recommendations made by online sites with

those made by friends for books and movies and reported that users prefer recommendations from close friends. They attributed this result to the assumption that friends would have similar preferences, as people tend to have close personal relationships with those who share their tastes and preferences. The preference for recommendations from close acquaintances is more pronounced for products particularly related to personal taste, such as movies or music. Bonhard and Sasse (2006) also argued that social closeness and taste overlap were important components in building effective recommender systems and suggested integrating social networking applications with recommender systems.

A conversational partner's social role can be identified by visual and verbal cues. Faces and gestures are the most predominant visual cues for the social role of an agent (Araujo, 2018; Kocaballi, Laranjo, & Coiera, 2019). As voice-based chatbot agents must depend only on verbal cues for playing a certain role, a language style is the most highlighted cue. In a study of social cues for conversational agents, Feine, Gnewuch, Morana, and Maedche (2019) stressed that the same verbal content can be expressed in many different styles. The social role of the agent defines linguistic style cues for delivering the same content. The verbal styles used in conversational interactions can further be classified based on language style. Although a formal language style involves the use of sophisticated words and expressions, an informal language style in conversation uses colloquial expressions and first and second person personal pronouns (i.e., I and You) (Jakic, Wagner, & Meyer, 2017).

Regardless of the social roles and language styles of the agents, the content of the message that they deliver may be identical. However, the persuasiveness of the message varies based on the characteristics of the social role or users' preference for the social role that the agent is playing. As such, the social role of a friend with an informal language style could be a heuristic cue for persuasion.

We hypothesize that users will apply stereotypes and social norms based on the cues of the agent. Furthermore, we assume that the recommendations from the agent who fulfills the social role of a friend will have higher possibilities of building a positive attitude toward the product.

H2. Recommendations from a voice agent with the social role of a friend will lead to a more positive attitude toward the product compared with recommendations from a secretary-type agent.

3.3. Involvement with the product

Persuasion mechanisms differ by level of involvement; thus, it is critical to understand the characteristics of the product and the user's involvement with the product. Product involvement refers to the customer's level of interest and motivation for a specific product (Bruwer & Huang, 2012). It has been argued that the level of involvement is related to the value system of the customer and different customers may perceive the value of the same product differently and form different levels of involvement (Shang, Chen, & Liao, 2006; Zaichkowsky, 1985). The level of involvement is dependent on each customer's motivation for a specific product recommended in a persuasive message and often measured on a continuous scale (Li & Liu, 2017).

However, this perspective of user characteristics has practical problems. For brands that must design the messages and conversation agent interactions before recommendation, it is very difficult to directly measure or indirectly predict each customer's involvement level for a specific product. Thus, we appropriately assume that the high–low level of involvement with the product is also determined by the attributes of the product, for example, price of the product, consumption motives, and search-experience goods.

If the product is expensive, people tend to perform more searches and be more careful when making decisions (Zaichkowsky, 1985). By contrast, if the product is cheap, people tend to have low involvement with the product. Similarly, people would be more involved in choosing a product that he or she will use for multiple years such as a car or a smartphone, compared with a product for one-time use. Consumption motive is another attribute. Sivaramakrishnan, Wan, and Tan (2007) investigated the impact of having an anthropomorphic agent on an e-commerce site under different consumption motives: utilitarian versus hedonic. They found that when a user had a utilitarian consumption motive, the attitude toward the product was lower when the agent was available. Additionally, for those users, interaction with an agent lowered the evaluation of the site as well. They explained that users were in information-seeking mode under the utilitarian consumption motive, whereas users with experiential consumption motive did not search for information as much. The availability of an agent was regarded as a peripheral aspect of the site and was neglected by users with a utilitarian consumption motive.

Additionally, the accessibility of objective information and active searches for information are two of the characteristics of the high-involvement state with a product. Search goods and experience goods are product types and depend on whether the consumer can access information on the quality of the product and evaluate it before purchasing (Nelson, 1970). Search goods are products for which a person can collect information on quality and price through a search. Consumers can evaluate the quality of a product and the options available to them prior to purchase. Evaluation of the product is objective because it is based on information. By contrast, experience goods are products for which it is difficult and costly to access information on quality before purchase and consumption. Evaluation of quality may require experiencing the product, such as tasting a wine; as such, evaluation tends to be subjective and depends on the individual's tastes and preferences. In the case of search goods, consumers conduct a search to gather detailed, objective information, and review and compare products before making a purchase decision. This endeavor requires cognitive effort and can thus be considered a high-involvement condition. By contrast, it is assumed that involvement would be low for experience goods because it is difficult to obtain objective information on quality, and evaluation depends on the experience after the purchase.

The moderating effect of involvement should be considered deliberately within the framework of HSM. The primary attribute that differentiates HSM from ELM is the proposition that heuristic and systematic processing can co-occur (Chang, 2004; Eagly & Chaiken, 1993). This effect is the called the "additivity effect," which allows the simultaneous assessment of the message content (e.g., personalized recommendation) and heuristic cues (e.g., the social role of a friend with an informal language style), when customers are strongly motivated and highly involved with the product. That is, the effect of the personalized message may be contingent on how involved the customer is with the product of interest. This moderating effect of involvement on personalization demonstrated that personalized messages generated a more favorable attitude in the condition of high involvement (Li & Liu, 2017). To verify the additivity effect of the dual processing model of a persuasive message and to extend the findings of the literature on web personalization into the voice-shopping context, the following hypotheses on the interaction effects between product involvement and systematic/heuristic processing modes are proposed and examined.

H3. There will be an interaction effect between personalization and product involvement on the attitude toward the product. The effect of personalization (i.e., personalized content of recommendation will generate a more favorable attitude toward the product in voice shopping) will be more salient when the customer is highly involved with the product than when lowly involved.

H4. There will be an interaction effect between social role and product involvement on the attitude toward the product. Voice agents with the social role of a friend with an informal language style will have a greater impact on the attitude toward the product in low-involvement conditions.

Based on the additivity effect of the HSM framework, a hypothesis of

a three-way interaction among personalization, social role, and product involvement could be proposed. That is, in a high-involvement condition, both personalized content in systematic processing mode and the social role of an agent in a heuristic processing mode would be assessed together to form a more positive attitude toward the product. Given the rarity of research findings on the interaction effect of social role and personalization, we proposed an explorative research question (RQ) on the three-way interaction effect.

RQ. Is there a three-way interaction effect among personalization, social role, and product involvement on the attitude toward the product?

4. Methods

4.1. Experiment design

Experiments were conducted using a 2 (personalization: personalized content vs. without personalized content) x 2 (social role: friend vs. secretary) x 2 (product involvement: high vs. low) factorial design. Participants were randomly assigned to 2 (personalization) x 2 (social role) by using a between-group design, and product involvement was implemented within-group. As participants were exposed to two very different product-purchasing scenarios (smartphone and video-on-demand [VOD]), it was reasonable to assume that the carryover effect would be minimal for product involvement. To minimize the order effect, half of the participants in each group experienced the high-involvement product scenario first, followed by the low-involvement scenario. The other half experienced the low-involvement product scenario first followed by the high-involvement scenario.

4.1.1. Voice chatbot agent

We used a text-to-speech (TTS) demo version available online (https://ttsdemo.com/) to replicate the experience of having a conversation with the agent. To control the possible effects of vocal characteristics (e.g., gender, pitch, speed), we used the same female voice character with the preset standard vocal effect for all conversation conditions.

4.1.2. Involvement with the product: product types

Two types of products were chosen to set a high versus low product involvement condition. For the high-involvement condition, a smartphone was used as the product; for low involvement, a VOD program for a TV series was used. The smartphone was used because it is an essential product that most people carry; additionally, it is relatively expensive (i.e., approximately USD 1000) and used for multiple years once purchased. People actively conduct searches and carefully compare models and features before purchasing.

By contrast, a VOD program is an experience good, for which it is difficult to evaluate the quality before making a purchase. The cost of a VOD program is cheap (i.e., approximately USD 1 per program) and the evaluation of its quality depends on personal preferences and tastes. Because of the characteristics of the product, it is difficult to compare one product with another or conduct a search on the quality of the product before purchasing; this means a lack of investment in cognitive efforts. We assumed that users would have low involvement with the VOD program.

4.1.3. Personalization of content

The process of personalization starts with obtaining information on the customer's preferences and is followed by a message-customization process tailored to these unique specifications (Kramer, Spolter-Weisfeld, & Thakkar, 2007). The personalized content of product recommendation messages can be devised by using several approaches. A customer's usage history, content filtering, and collaborative filtering have been the most-used methods. We assumed that the AI-enabled voice agent had a customer's preference information on the product ahead of the event of recommendation because of the customer's usage history data.

As suggested by a study on the personalization of online ads (Li & Liu, 2017), we designed the personalized recommendation message to generate a match–mismatch condition. To prepare customized content scripts before the experiment, a few days prior to the experiment, participants assigned to the personalized content treatment were asked to answer a pre-survey on their usages and preferences with demographics. For smartphones, we asked what features or apps they used frequently and what they considered important features. Based on the answers, we prepared customized scripts for each participant. The customized script included references to features the person used the most, favorite applications, merits of the recommended product specific to the feature, and reviews from other users in the same gender and age group. To control possible confounding effects from other content-related issues, the number of product attributes and the length of recommendation script were set to be consistent with the personalization and non-personalization treatment.

For example, for a female aged in her 20s who reported her top three preferences of photos, Instagram, and display size, a personalized recommendation script of a secretary-role agent was designed as follows: "I would like to recommend iPhoneX, as you used the camera the most, and it comes equipped with 12 MP dual camera. It is recommended by females in their 20s who frequently use Instagram and photo apps. It has a 5.8-inch super retina display that exactly meets your preferences."

By contrast, for the non-personalization group, general product descriptions from the official product website were provided. For example, the iPhoneX's top three features listed on the brand promotion webpage were transcribed into a recommendation content as follows: "I would like to recommend iPhoneX. First, it provides a 5.8-inch super retina display. Second, it has a 12 MP dual camera, which lets you capture studio-quality photos. Lastly, for advanced and secure authentication, the Face ID feature is supported."

In this case, the Face ID, a new feature highlighted by the brand's product website, was excluded from the personalized-content script, and exposure order of other features was set differently in the order of camera, photo apps, and display size for each personalized-content script.

The format of personalized-content design for the VOD condition was similar to that aforementioned. For participants assigned to the VOD program group, in the pre-survey, we asked what programs they had watched recently, which TV series they followed, and what their favorite programs were. The personalized script included references to recently watched or favorite programs, recommendations from people who watched the same programs, and a brief introduction to each recommended program. For the non-personalization group, we recommended a program from among the top three in the previous week's TV program ranking, and the program description was taken from the official program website.

4.1.4. Social roles of a secretary and a friend

Social roles were differentiated by tone and words used. The secretary-role conversational agent used official, formal, and professional words. For the friend role, words commonly used in daily conversations with close friends were used, in a casual, informal manner. The secretary-type used an official tone such as "would like to," but the friend-type used a casual tone like "how about" and called the participant "you," to generate a feeling of a more personal relationship.

For example, the secretary-role agent would say, "I would like to recommend iPhoneX. It is recommended by females in their 20s who frequently use Instagram and photo apps. It has a 5.8-inch super retina display, which exactly meets your preferences," whereas the friend-role agent would say "How about iPhoneX? Women your age who are into Instagram and photos gave two thumbs up. The 5.8-inch super retina display is just for you." The same format of content design differentiated

by the secretary or friend roles was applied to the VOD content. Four versions of the scripts for a smartphone recommendation are presented in Table 1.

4.2. Measure of the dependent variable

As a dependent variable, the attitude toward the product was adopted because it is a precondition for purchase decisions and a well-established index for experimental studies (Stuart, Shimp, & Engle, 1987). Three sematic differential items from Petty et al. (1983) were used to measure the attitude toward the product. Participants were asked to rate their impression of the product (bad/good, unsatisfactory/satisfactory, unfavorable/favorable) on a seven-point scale (−3 to +3).

4.3. Manipulation check of treatment condition variables

Pilot tests were conducted with 36 participants to verify the manipulation. All treatment conditions were confirmed as properly manipulated. All items were measured on seven-point Likert scales with 1 = *strongly disagree* and 7 = *strongly agree*. To verify the manipulation for social role, we developed three items: 1) The tone and manner of the agent were suitable between friends, 2) The conversation I had with the agent was something I would have with my friend, and 3) It felt like the agent was my friend. The reliability of the three-item construct was 0.88. There was a significant difference ($t = 5.11$, $p < 0.01$) in how participants felt about and recognized the social role of the agent between the group who communicated with the friend-type agent ($M = 4.29$, $SD = 0.84$) and the group who communicated with the secretary-type agent ($M = 3.00$, $SD = 0.66$).

As pointed out by Li (2016), perceived personalization should be distinguished from actual personalization and thus a manipulation check of a participant's perception is required. Personalization manipulation was tested by using items from Tam and Ho (2005). The results of the *t*-test showed a significant difference ($t = 2.72$, $p < 0.01$) between the group who received personalized recommendations ($M = 5.02$, $SD = 1.53$) and the group who received recommendations without personalization ($M = 3.80$, $SD = 1.07$). Items from the Personal Involvement Inventory (PII) scale developed by Zaichkowsky (1985) were used to measure product involvement. Participants who experienced the smartphone scenario showed higher involvement ($M = 1.04$, $SD = 1.09$) compared with those who experienced the VOD scenario ($M = 0.34$, $SD = 0.72$; $t = 2.23$, $p < 0.05$).

4.4. Participants

A total of 124 undergraduate and graduate students at a university in an urban area were recruited. The student sample was used because they are active users of smartphones and VOD services, and familiar with conversational agents. After removing two answers for which values were missing, we analyzed data from 122 participants (53% male and 47% female: 89% were aged in their 20s, and 11% in their 30s). Each participant received a gift card worth USD 8 for participating. Each session lasted approximately 30 min. Participants assigned to the personalized message group were asked to answer a pre-survey on their usage and preferences.

4.5. Experimental procedure

The experiment was conducted one person at a time in a laboratory at the university. Each participant was asked to sit in front of a desk with a PC monitor and a speaker. On the PC screen was an explanation of the experimental scenario that the participant was to follow. The participants were informed that the speaker was AI-enabled, and the voice agent would recommend a product after exchanging some questions and answers.

To control a pre-known brand effect (e.g., Amazon, Google), a plain shaped, Bluetooth-enabled speaker with a logo and the brand name masked was used to replicate the experience of interacting with a conversational voice agent. As illustrated in Fig. 1, the speaker was placed on the participant's desk, connected to a researcher's computer in another room. The researcher entered each prewritten script of questions and recommendation messages according to the participant's answer, which was played through the speaker through TTS in the participant's room.

The overall experimental procedure is illustrated in the flowchart diagram of Fig. 2. For the high-involvement condition, the participants were given a scenario in which they were thinking of buying a new smartphone. According to the scenario, before going to offline stores to check the models, they decided to perform some research. Each experiment comprised two parts: a preformatted conversation between the participant and the agent, and the agent's recommendation message for the product

In the beginning, each participant was asked to activate the conversational agent by calling its name, Jenny. Once activated, the agent introduced herself with a greeting ("Hello, I am your assistant, Jenny," for the role of a secretary, or "Hello, my friend. I am Jenny," for the role of a friend). Next, the agent asked a few questions including what product the person was looking for (participants were to answer, "I am looking for a new smartphone.") and briefly described the market price, which was approximately USD 1000. Next, the agent asked whether they wanted Android or iOS (participants were to answer, "I want an Android phone," or "I want an iOS phone").

Upon the assignment of personalization, the researcher in another room played a preset script of a recommendation message. For example, if a participant was assigned a personalized content–friend role condition and wanted an iOS phone, the voice agent recommended the iPhone X with product attributes based on the participant's preferences, with

Table 1
Sample script of social role and personalization treatment (product: iPhoneX).

	Secretary	Friend
Without Personalization	I *would like to recommend* iPhoneX. First, it provides 5.8-inch super retina display. Second, it has 12 MP dual cameras, which will let you take studio quality photos. Lastly, for advanced and secure authentication, *the Face ID feature is supported.*	*How about* iPhoneX? *You* will like the 5.8-inch super retina display. *You* can take studio quality photos with 12 MP dual cameras. For secure unlocking, try the new Face ID. *Your face is your password.*
With Personalization	I *would like to recommend* iPhoneX. As you used the camera the most, and it comes equipped with 12 MP dual camera. *It is recommended* by females in their 20s who frequently use Instagram and photo apps. It has a 5.8-inch super retina display that exactly meets your preferences.	*How about* iPhoneX? You used the camera every day and it comes equipped with 12 MP dual camera. *Women in your age* who are into Instagram and photos *gave two thumbs up.* The 5.8-inch super retina display is *just for you.*

*Phrases *in italics* show differences in language style between a secretary social role and a friend role.
**Phrases that are underlined illustrate personalized scripts, which include an explanation of the participant's favorite app and the rationale behind the recommendation based on the pre-survey performed before the experiment.

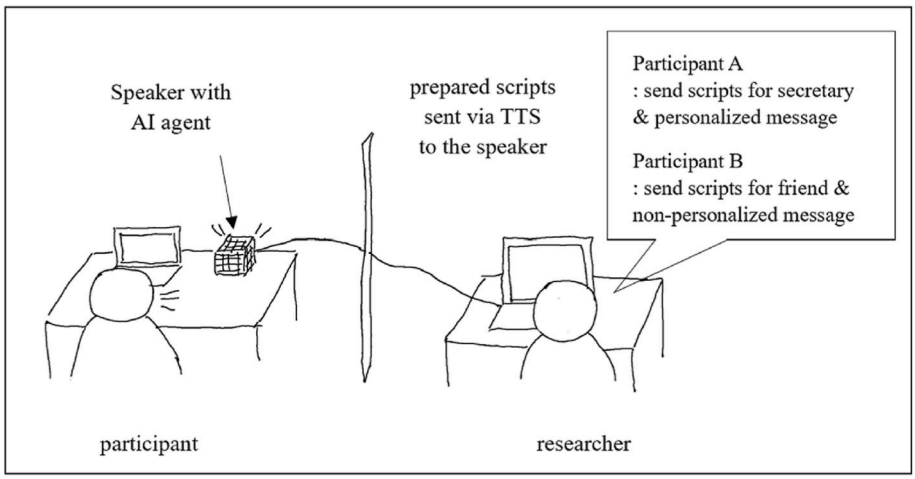

Fig. 1. Experimental setup.

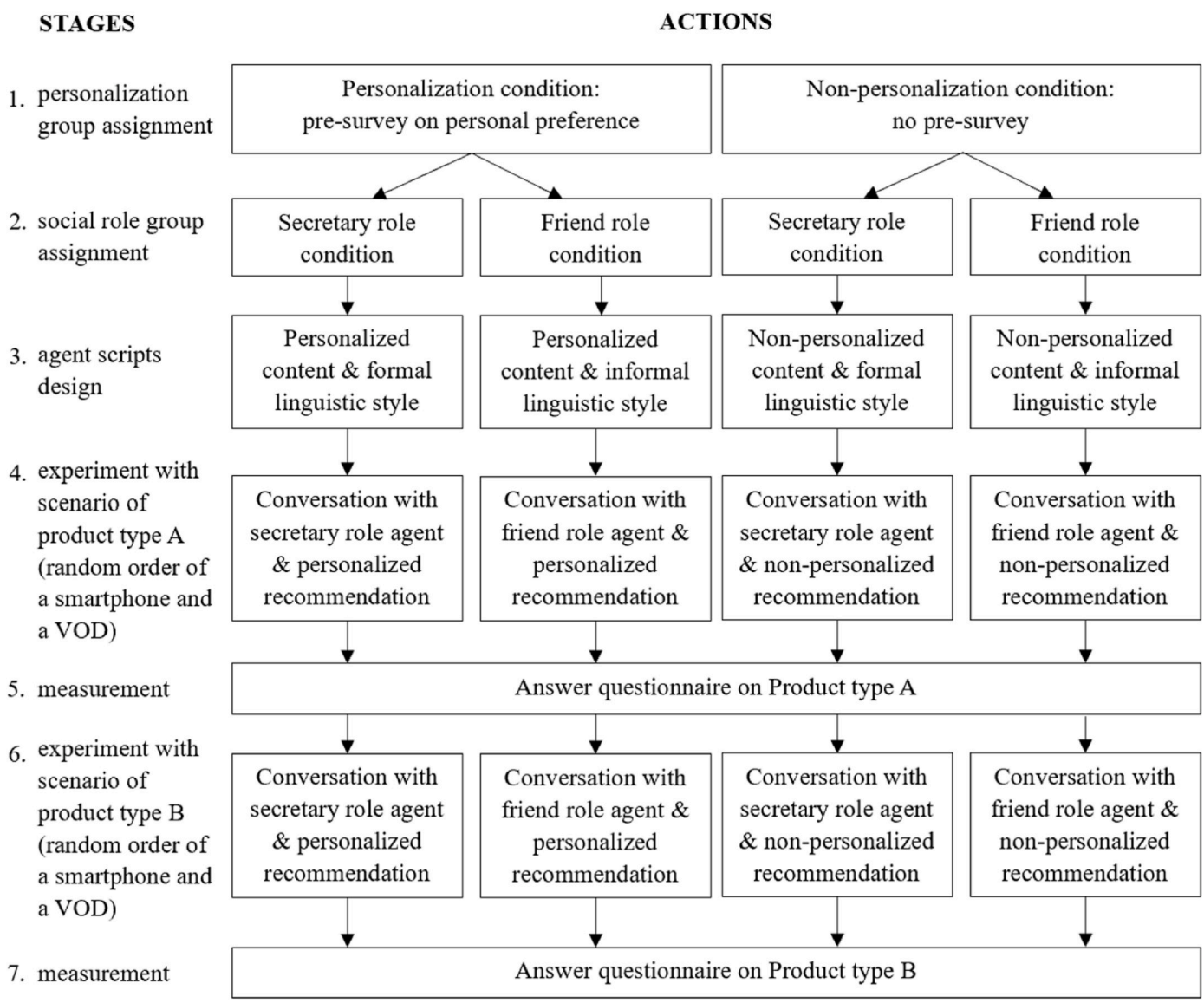

Fig. 2. Flowchart of experiment.

Table 2
Means and standard deviations of attitude toward the product.

	Low Involvement (n = 122)		High Involvement (n = 122)	
	Without Personalized Content (n = 61)	With Personalized Content (n = 61)	Without Personalized Content (n = 60)	With Personalized Content (n = 62)
Secretary Role (n = 62)	0.35 (1.55) (n = 31)	1.16 (1.20) (n = 31)	0.44 (1.40) (n = 30)	1.41 (1.03) (n = 32)
Friend Role (n = 60)	1.43 (0.91) (n = 30)	1.56 (1.14) (n = 30)	0.43 (1.32) (n = 30)	1.60 (0.87) (n = 30)

Note: Attitude toward the product was measured with seven-point sematic differential scales anchored at −3 to +3. n = number of participants per each condition and assigned group.

Table 3
Results of three-way ANOVA with attitude toward product as a dependent variable.

Source	Type III Sum of Squares	D.F	F	p	Partial η2
Personalization	35.61	1	24.72**	0.000	0.095
Social Role	10.44	1	7.25**	0.008	0.030
Involvement	1.47	1	1.02	0.314	0.004
Interaction effects					
Personalization x Social Role	0.88	1	0.61	0.436	0.003
Personalization x Involvement	5.49	1	3.81	0.052	0.016
Social Role x Involvement	6.34	1	4.40*	0.037	0.018
Personalization x Social Role x Involvement	3.01	1	2.09	0.150	0.009
Error	339.98	236			

*p < 0.05, **p < 0.01.

informally styled words and expressions of a friend. For participants in the non-personalized content–secretary role group, the voice agent recommended the iPhone X with general product attributes highlighted on the brand's website with a formal language style of a personal assistant.

For the low-involvement condition, participants were informed that it was 9 p.m. on a Sunday. Participants were asked to activate the agent by calling its name and checking what was on TV now. After exchanging a few questions and answers on what was being played on live TV channels, they talked about an alternative option, the VOD program. The agent briefly described the price, which was approximately USD 1 per program, and asked whether the user would prefer drama or a variety of entertainment shows. As in the high-involvement condition, scripts were prewritten for two options, and the agent provided recommendations based on the experimental conditions of the personalized content and the social role of the voice agent. Upon completion of each scenario, the participants responded to the questionnaires on the attitude toward the product recommended by the voice agent.

5. Results

We analyzed data from 122 participants. Because the product involvement condition was conducted with within-group design, 244 cases were input into statistical analysis by using SPSS version 25.0. Instead of a regression model, an analysis of variance (ANOVA) test was used for analysis because all independent variables were two-level categorical scales.

Approximately 30–32 cases were randomly assigned in each of the eight condition groups, with almost an equal ratio of gender and age in the groups. The descriptive statistics for attitude toward product are summarized in Table 2. The mean scores of attitude toward product (range: -3 to +3) were highest for the high-involvement (smartphone)–personalized content–friend role group ($M = 1.60$, $SD = 0.87$), and the lowest scores were for the low-involvement (VOD)–non-personalized content–secretary role group ($M = 0.35$, $SD = 1.55$).

To verify the main and interaction effects on the attitude toward product, a three-way ANOVA was conducted with personalization, social role, and involvement as explanatory variables (Table 3). The results showed significant main effects for personalization (F (1, 236) = 24.72, $p < 0.01$) and social role of a friend (F (1, 236) = 7.25, $p < 0.01$). Participants formed a more positive attitude toward products when a personalized recommendation was provided ($M = 1.43$, SD = 1.07) compared with a recommendation without personalization ($M = 0.66$, SD = 1.38). Additionally, participants liked the product more when it was recommended by a conversational agent with the role of a friend ($M = 1.26$, SD = 1.17), rather than by the agent with the role of personal assistant ($M = 0.85$, SD = 1.37). According to the ELM, people tend to be more skeptical under high-involvement conditions because they

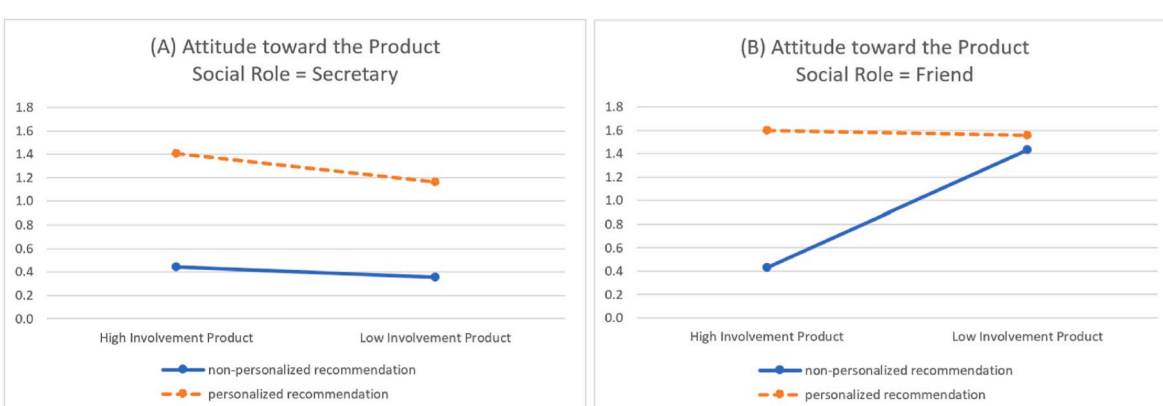

Fig. 3. Interaction effects for attitude toward the product. Note. Panel (A) shows the interaction effect of personalization and product involvement on the attitude toward the product for the secretary-type social role. Panel (B) shows the interaction effect of personalization and product involvement on the attitude toward the product for the friend-type social role.

scrutinize and evaluate the product's quality in depth. However, in this research, there was no main effect of involvement on the user's attitude.

We observed an interaction effect of role x involvement (F (1, 236) = 4.40, $p < 0.05$). Social role had a significant impact on attitude toward product under low-involvement conditions (F (1, 236) = 11.48, $p < 0.01$) but not under high-involvement conditions (see Fig. 3). These results are consistent with the prediction of ELM: the peripheral route is used under low-involvement conditions.

The personalization × involvement interaction did not reach significance. The interaction effect of personalization x role and personalization x role x involvement were not significant as well.

6. Discussion and implications

6.1. Summary of findings and discussion

This study examined the persuasion mechanism in product recommendations made by a voice-based conversational agent and explored whether the personalized content and the agent's social role of a friend, rather than a personal assistant, generated a more positive attitude toward the product in the context of voice shopping. A set of hypotheses and a research question were proposed to identify the effects of personalization and social role, and product involvement, on the building of a positive attitude toward the recommended product. Personalized content of the message, and a friend role for the agent with an informal language style, were adopted as central and peripheral routes, and involvement with the product was set as a mediating condition.

As hypothesized in H1 and H2, both personalization and social role had significant impacts on the attitude toward the product. In the experiment, the script of personalized content was devised based on each participant's unique preferences for product attributes. Personalized content had a strong impact and led to positive attitudes. Attitudes formed by personalized recommendations were substantially more positive than attitudes formed by non-personalized content in all conditions, and the difference between the two groups was substantial, except for the low-involvement/friend role scenario. These results indicate that personalized recommendation is a key feature for voice shopping. This finding is consistent with studies of the effects of personalized messages on the web (Ho & Bodoff, 2014; Li, 2016; Li & Liu, 2017; Tam & Ho, 2005). As pointed out by Li (2016) and Li and Liu (2017), the inconsistent effect of personalization in the prior studies might be generated by multiple approaches to personalization. We found a very strong effect of personalization when a customer's preferences for product attributes are reflected in the content of the persuasive message.

The results also supported the effect of a friend role of a voice agent on the attitude toward the product (H2). Participants liked the product more when it was recommended by a conversational agent with the role of a friend who used an informal language style, rather than by the agent with the role of personal assistant or a secretary who used formal language style. Although studies on this subject have been rare, this finding is consistent with the effect of a friend-role agent in the recommender system of online sites (Bonhard & Sasse, 2006; Sinha & Swearingen, 2001).

Additionally, as posited in the dual processing models of persuasion, social role had a greater impact under low-involvement conditions. As proposed in H4, the interaction effect between social role and product involvement was significant and supported. This finding of the mediating effect of product involvement on a voice agent's social role has not been reported in the literature. The peripheral route and heuristic cues of the informal language style of a voice agent played a bigger role under low-involvement than in the high-involvement condition. When a user had high involvement with the product, the quality of content and the level of personalization were more important, and who delivered the message did not seem to matter much for persuasion. However, under the low-involvement condition, by whom the message was delivered and the sense of closeness with the agent affected attitude formation.

The only hypothesis not supported was H3, which proposed the interaction of personalization x involvement. Personalized recommendation led to favorable attitudes under both conditions of high and low product involvement. Even under the low-involvement condition, the attitude formed by the personalized recommendation from the secretary-role agent was much more favorable than the attitude formed by the recommendation without personalization from the same role of agent. Thus, it can be interpreted that, when a customer's preferences for product attributes are highly reflected in the content of a recommendation, the customer is affected by the personalized content regardless of the levels of involvement with the product.

The three-way interaction effect of RQ was not found. The additivity effect of the HSM framework claims that the quality of messages and heuristic cues can co-occur when the customer has a strong motivation in a high-involvement condition (Chen & Chaiken, 1999). In a high-involvement condition, both personalized content and the social role of an agent would be processed together to form a more positive attitude toward the product. Li and Liu (2017) found this additivity effect in their study of personalized greeting messages in online ads, but our study of personalized content formed by individual preferences did not verify the effect in the voice-shopping context. In the results of the means of the treatment groups (Table 2), the personalization friend-role treatment groups had the highest score for attitude toward the product both in high- and low-involvement conditions. Additionally, the results of the ANOVA test demonstrated that personalization had the strongest explanatory power in predicting the attitude change, but no interaction effect with involvement and social role was observed. Thus, it can be interpreted that high-depth personalized content, such as applied in this study, affects the customer's attitude regardless of the agent's characteristics or product types.

6.2. Theoretical implications

As proven in many marketing, advertisement, and information systems studies, ELM and HSM are useful tools for understanding the effectiveness of communication and persuasion mechanisms in interactions with a conversational agent. Although few studies on conversational agents have applied this persuasion perspective, it addresses attitude formation from a dual-process perspective and can be used to understand why one route of the dual processes would be predominant in a particular case. Heightened personal relevance through personalized messages can be a central route/systematic processing strategy to promote high involvement with a product and encourage cognitive efforts regarding the message delivered. Also, enriching a social trait of the conversation agent serves as a peripheral route/heuristic processing strategy when the user does not have a high level of involvement with the product.

One of the research design limitations of the dual processing model is the manipulation and measurement of product involvement level. In a post hoc analysis, we further analyzed the data with the product involvement variable on a Likert-type seven-point continuous scale, such as adopted in the literature (Huang & Zhou, 2018; Li & Liu, 2017). A test of the regression model showed results identical to those of the ANOVA test, which put product involvement on a categorical scale (1 = low-involvement product, 2 = high-involvement product). The interaction effect was not found between personalization and product involvement but was found to be significant between social role and product involvement ($\beta = 1.48$, $p < 0.01$). The three-way interaction was not significant. Thus, we found that the different approaches to the manipulation of product involvement generated the same results in tests of the effect of personalization (systematic processing) and social role (heuristic processing).

This research contributes to the literature by examining the persuasion mechanism of conversational voice agents. The results

reinforce the CASA paradigm and show that the tone and words used during the conversation trigger social cues regarding the agent's social role. Various studies have investigated the impact of the gender and personality of the agent, but few have explored the effects of their social role. In this study, effects of social role were examined in the context of product recommendation. Similar to how people rely more on recommendations from close friends in real life, a friend-role agent was more effective in inducing favorable attitudes compared with a secretary-role agent.

6.3. Practical implications

The research findings highlight several points that need to be considered during the interaction design and product-planning phase for the conversational agents for voice shopping. Many conversational agents implicitly follow the social role of personal assistant because the first and most-used agents tended to be promoted as digital personal assistants. As such, most agents have the tone and manner of a personal assistant; they are friendly, but in a polite, formal manner. The tone and words used are different from those used in daily conversations among close friends. Murray and Häubl (2009) suggested that there could be a correlation between the "effectiveness" and "affectiveness" of the conversational agent. During the product and marketing planning phase, user–agent interaction scenarios must be thoroughly reviewed, including the product types that the agent will cover and where the product will be integrated into. Based on the context, the persona of the conversational agent must be designed to fit the conversation and product recommendation setting. The alternative social role of conversational agents may be applied to different practical contexts, such as counseling, education, crowdsourcing, virtual environment, and telerehabilitation (Nakayama et al, 2018, 2019).

The data show that users recognize social cues from the tone and words used and apply the social role triggered by the cue to the agent. When preparing scripts, the tone and words must be chosen strategically to successfully represent the desired social role. With the rapid growth of voice shopping, marketers are already preparing for voice search engine optimization (Bentahar, 2018). How the snippets of product information are delivered through conversational agents and which tone and words are used could affect attitudes toward the product and eventually purchase intentions.

Another important finding was the critical role of personalization on attitude change. Personalization starts with analyzing users' behavioral patterns and the data of others who have similar interests or needs. Agent features, interactions, and data collection must be carefully planned while considering personalization. One of the biggest challenges of personalization is customer's concerns for privacy (Montgomery & Smith, 2009). With rising concerns about privacy and regulations, such as the European Union's General Data Protection Regulation, more thorough consideration and planning is required regarding which personal data to collect and store, how to perform the collection, and how to obtain consent from the user.

6.4. Limitations and recommendations for further research

This study attempted to understand the persuasion mechanism of a recommendation voice agent, and the main focus was on personalization and social role. As persuasion is not a simple process but a complex mechanism affected by various factors, this study has several limitations that can inspire further research.

Regarding the level of involvement, we assumed that the search goods would have higher involvement compared with the experience goods because the consumer can access information on the quality and tends to perform more search and scrutinization. We also considered the price and duration of the usage when choosing the product for the experiment because those are key aspects that affect involvement. As a single product may have multiple facets and various usage patterns, it may be difficult to classify products as a high-involvement or low-involvement product, based on one or two characteristics such as price range, usage time, or product experience. For example, when planning for a vacation and booking a hotel, the price for the hotel reservation could be high but used for a short period of time. Additionally, information on hotels is easily accessible from booking sites, but there are limits to evaluations of actual quality because they depends on experience and personal preference; for this reason, people often rely on comments and ratings from customers who have stayed there before. Likewise, it is difficult to categorize and define the level of involvement for certain types of products.

More studies on voice shopping through conversational agents are necessary to clarify the effects of product characteristics and involvement level with products. Although we focused on the price and use duration of the product in this research, many other factors affect the level of involvement. Zaichkowsky (1985) classified these factors into three categories: personal, physical, and situational. The same product could result in varying levels of involvement among different people and different purchase situations. In particular, for a conversational agent, the user's level of involvement varies based on who initiated the dialogue. For example, the involvement level would be much higher when the user initiates a conversation with an agent through their smartphone to ask about different brands of milk and their prices, compared with the case where an agent integrated into the refrigerator initiates the dialogue and recommends a type of milk. As agents are increasingly being integrated into home appliances, understanding how situational factors impact involvement and overall persuasion mechanisms will become more important. Lastly, to generalize the results of this study, a more diverse age group should be included in the sample for further research.

CRediT authorship contribution statement

Chong Eun Rhee: Conceptualization, Writing - review & editing, Formal analysis, Data curation, Writing - original draft. **Junho Choi:** Conceptualization, Writing - review & editing, Supervision, Formal analysis, Validation.

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